

Cairo University

Faculty of Computers and Artificial Intelligence
Third Year — Data Warehouse Project

Brazilian E-Commerce Data Warehouse

Based on the Olist Brazilian E-Commerce Public Dataset

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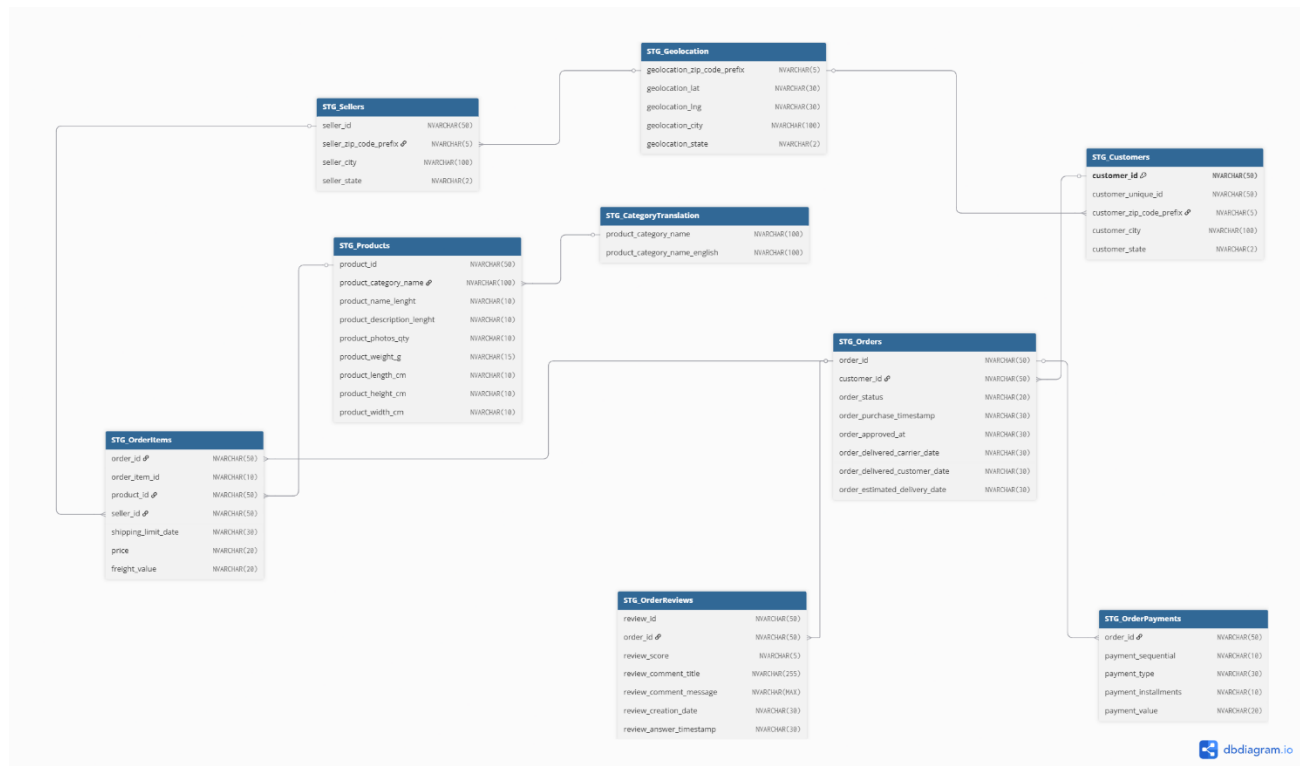
1. Physical Model of the Source System

The source system is the Olist Brazilian E-Commerce public dataset available on Kaggle. It consists of nine interrelated CSV files that together represent a Brazilian marketplace where multiple sellers fulfill orders placed by customers across Brazil.

The nine source tables are:

- olist_customers_dataset — customer identity and location
- olist_geolocation_dataset — ZIP code-to-latitude/longitude mapping
- olist_order_items_dataset — individual line items within each order
- olist_order_payments_dataset — payment details per order
- olist_order_reviews_dataset — customer reviews and scores per order
- olist_orders_dataset — order header with status and timestamps
- olist_products_dataset — product catalog with category and physical attributes
- olist_sellers_dataset — seller identity and location
- product_category_name_translation — Portuguese-to-English category name mapping

The physical diagram below shows the staging (STG) tables that mirror these CSV files, including data types and relationships:



2. Dimensional Model

2a. Business Processes

Three core business processes were identified and modeled:

- Order Item Fulfillment — tracks individual products sold within each order, capturing price, freight, seller, and product information.
- Payment Processing — tracks all payment transactions associated with orders, including payment method, installments, and value.
- Customer Reviews — tracks customer satisfaction through review scores submitted after order delivery.

2b. Grain of Each Fact Table

Fact Table	Grain (one row represents...)
Fact_Order_Items	One line item within one order — a single product sold by a single seller in a given order.
Fact_Payments	One payment record for one order — a single payment installment/type associated with an order.
Fact_Reviews	One customer review for one order — the review score and creation date submitted by the customer.

2c. Type of Each Fact Table

Fact Table	Type	Explanation
Fact_Order_Items	Transactional	Each row is a discrete transaction event — an item purchased. Rows are never updated.
Fact_Payments	Transactional	Each row records a payment event. Multiple payment types per order are stored as separate rows.
Fact_Reviews	Transactional	Each row records a review submission event tied to a single completed order.

2d. Dimensions and Their Types

Dimension	Type	Description
Dim_Customer	SCD Type 2	Stores customer identity and location. Supports historical tracking via effective_start_date, effective_end_date, and is_current flag.
Dim_Geolocation	Static (Type 0)	Maps ZIP code prefixes to latitude, longitude, city, and state. Surrogate key (geo_sk) used because ZIP codes have duplicate entries in the source.
Dim_Orders	SCD Type 1	Stores order header information including status and all timestamps. Acts as a conformed dimension shared by all three fact tables.
Dim_Product	SCD Type 1	Stores product ID and category name. Category name links to Dim_Category_Translation for English translations.
Dim_Sellers	SCD Type 1	Stores seller identity and ZIP code prefix. State is accessible via Dim_Geolocation join at query time.
Dim_Category_Translation	Static (Type 0)	Maps Portuguese product category names to their English equivalents. Must be fully loaded before Dim_Product.

Note: The minimum requirement of 6 dimensions is met: Dim_Customer, Dim_Geolocation, Dim_Orders, Dim_Product, Dim_Sellers, and Dim_Category_Translation.

2e. Measures in Fact Tables

Fact_Order_Items

Measure	Data Type	Type	Description
price	DECIMAL(10,2)	Additive	Sale price of the individual product unit.
freight_value	DECIMAL(10,2)	Additive	Freight/shipping cost charged for that item.
shipping_limit_date	DATETIME	Non-additive	Deadline for the seller to hand off to carrier (degenerate dimension-like attribute).

Fact_Payments

Measure	Data Type	Type	Description
payment_value	DECIMAL(10,2)	Additive	Monetary value of the payment installment.
payment_installments	INT	Semi-additive	Number of installments chosen by the customer.

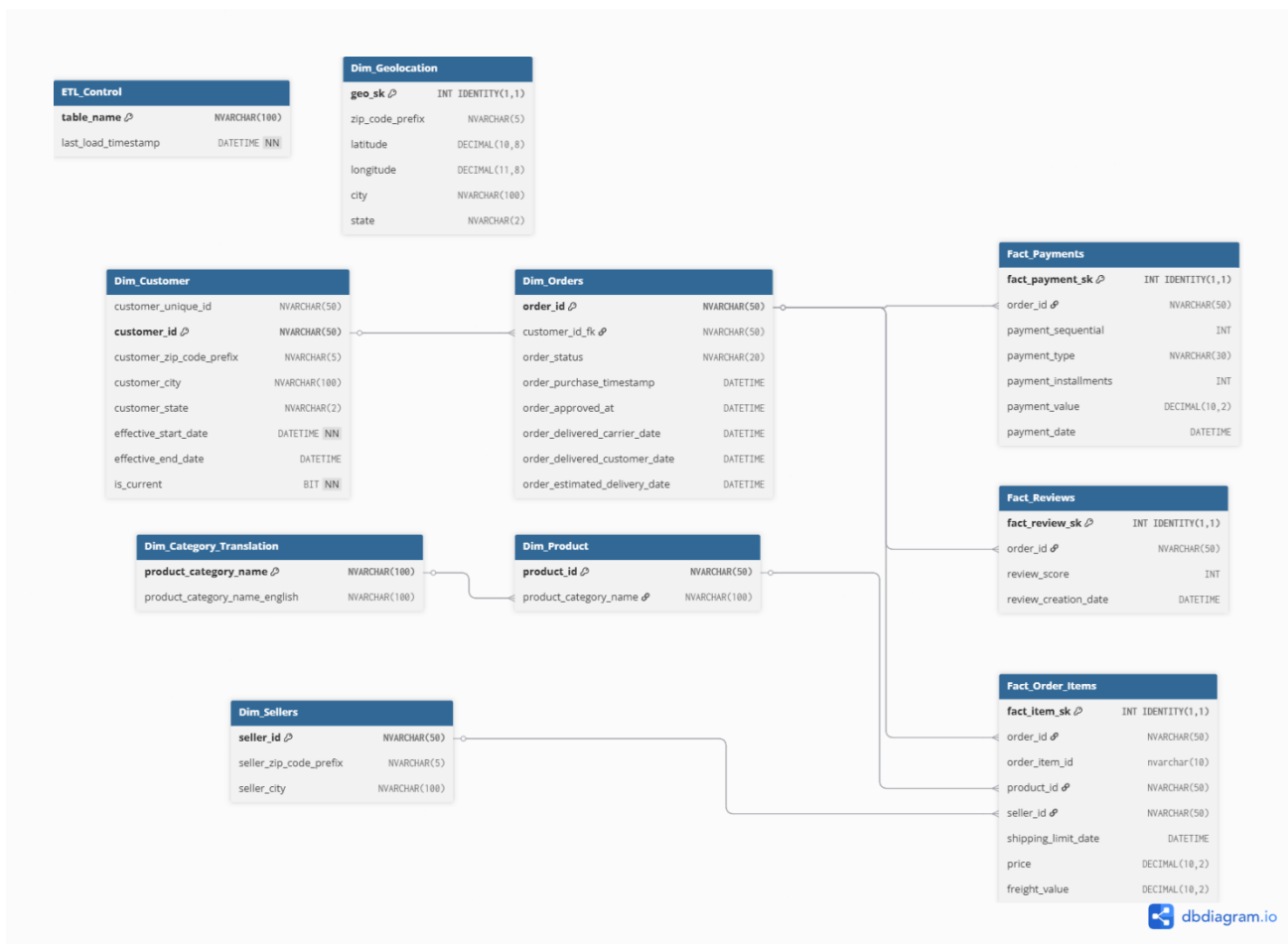
Measure	Data Type	Type	Description
payment_sequential	INT	Non-additive	Sequence number of the payment row within an order.
payment_date	DATETIME	Non-additive	Date derived from order_purchase_timestamp via lookup on Dim_Orders during ETL.

Fact_Reviews

Measure	Data Type	Type	Description
review_score	INT	Semi-additive	Customer satisfaction score (1–5). Averages are meaningful; sums are not.
review_creation_date	DATETIME	Non-additive	Timestamp when the review was created by the customer.

2f. Star/Galaxy Schema

The dimensional model follows a galaxy schema (multiple fact tables sharing conformed dimensions). The central conformed dimension is Dim_Orders, referenced by all three fact tables.



3. ETL — Data Flow and Control Flow Tasks

Two SSIS packages were developed:

- Load_STG.dtsx — Extracts data from the nine CSV source files and loads them into the staging database (STG_Brazilian_DW).
- Load_DW.dtsx — Transforms and loads data from staging into the data warehouse (Brazilian_DW), including SCD Type 2 logic for Dim_Customer and incremental loading via the ETL_Control watermark table.

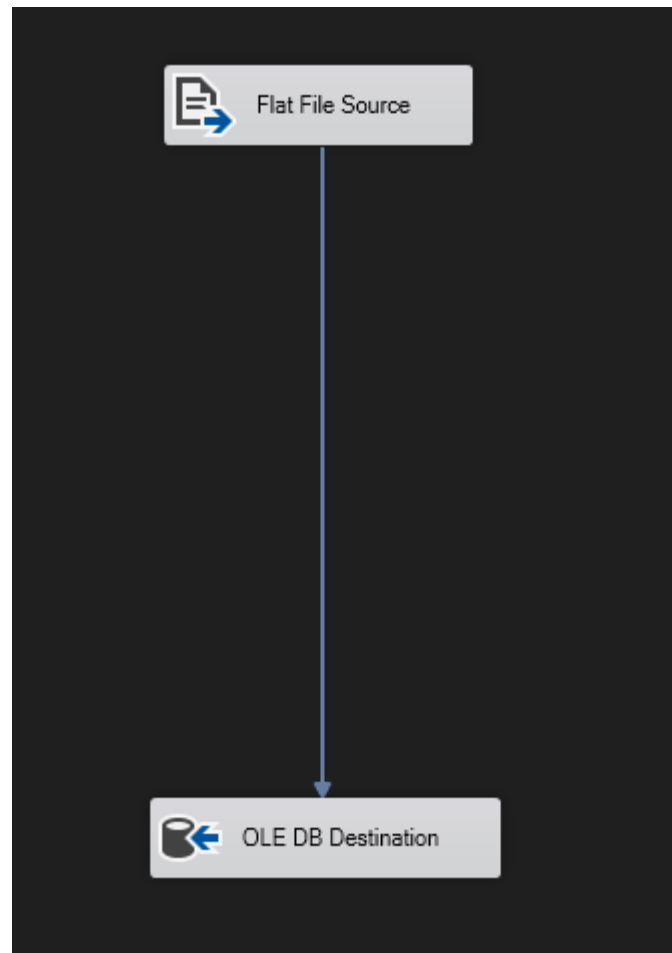
3.1 Load_STG Package — Control Flow

 Truncate STG tables

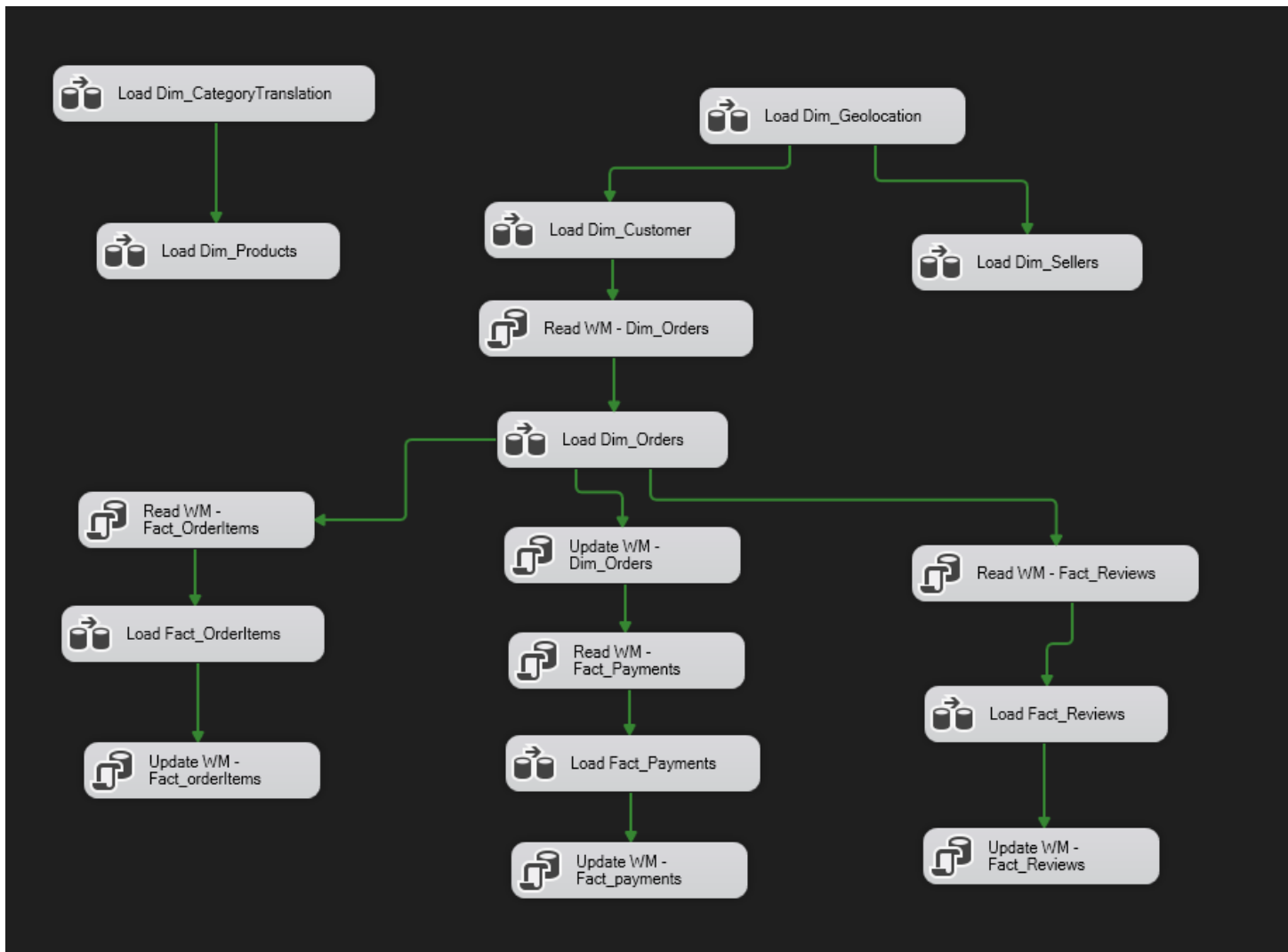
 Load CSV

-  Customer_STG
-  Geolocation_STG
-  OrderItems_STG
-  OrderPayments_STG
-  OrderReviews_STG
-  Orders_STG
-  Products_STG
-  Sellers_STG
-  CategoryTranslation_STG

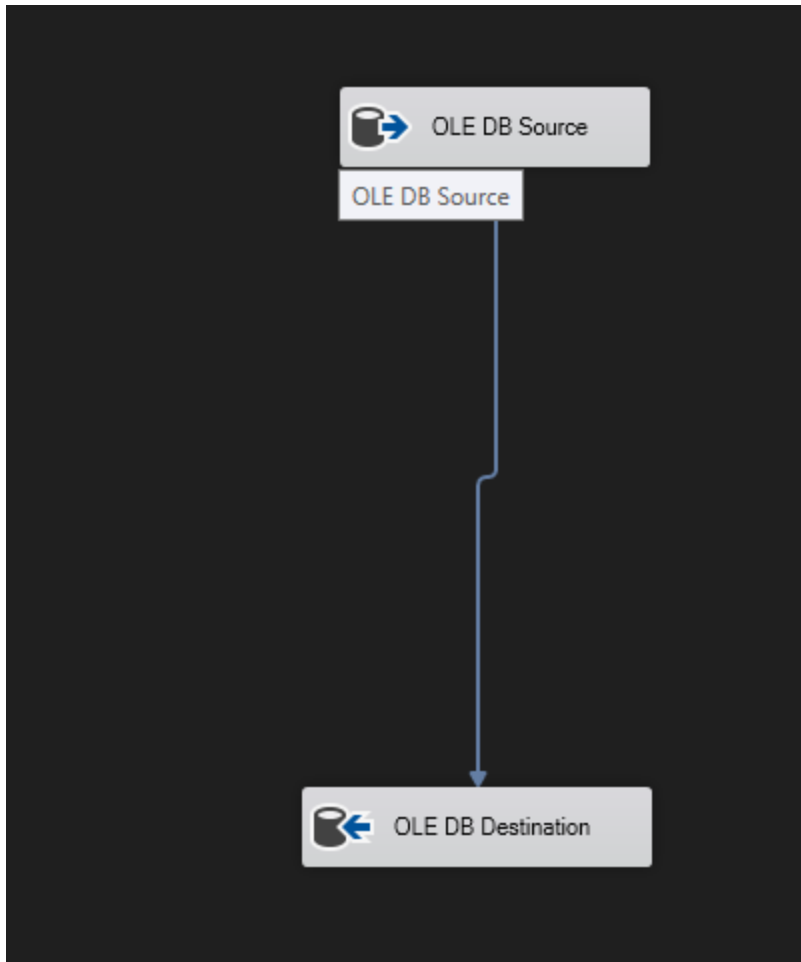
3.2 Load_STG Package — Data Flow: Customers, Orders, Order Items, Order Payment, Order Reviews, Products, Sellers, Category Translation & GeoLocation



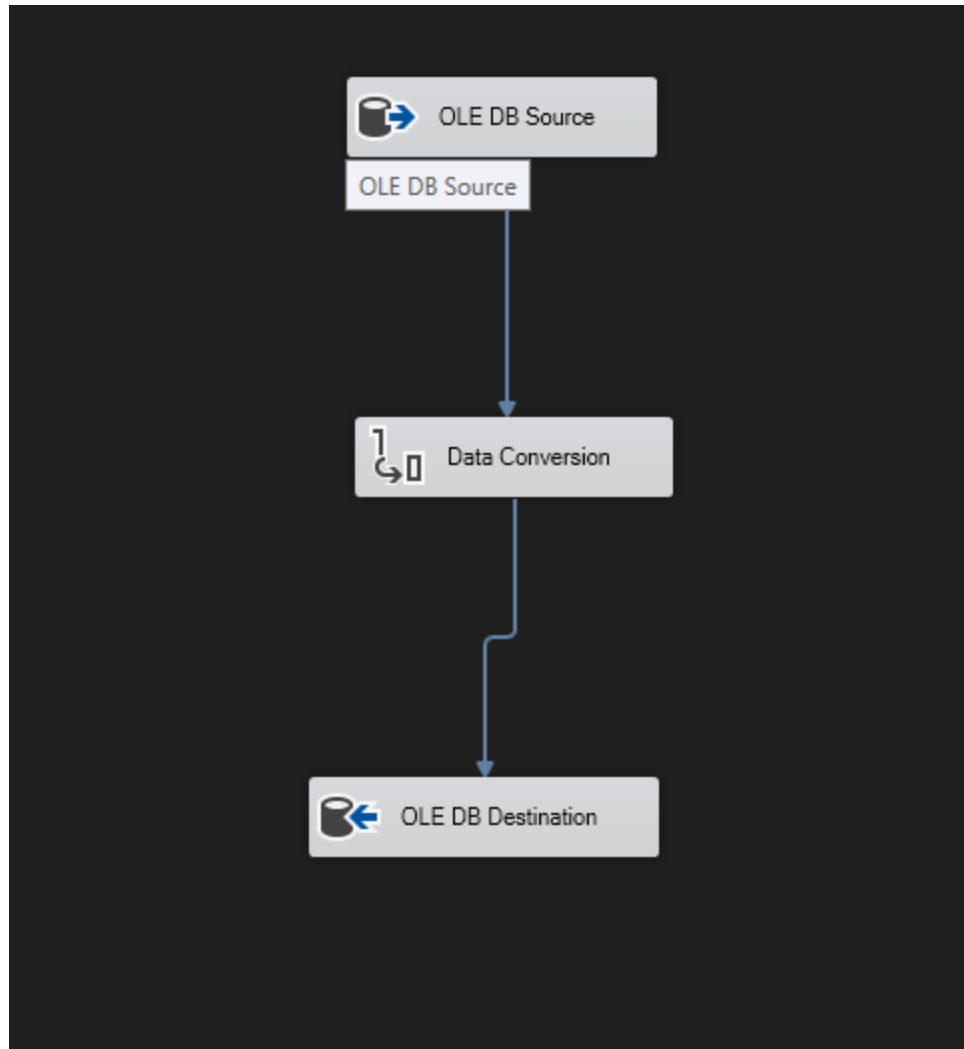
3.9 Load_DW Package — Control Flow



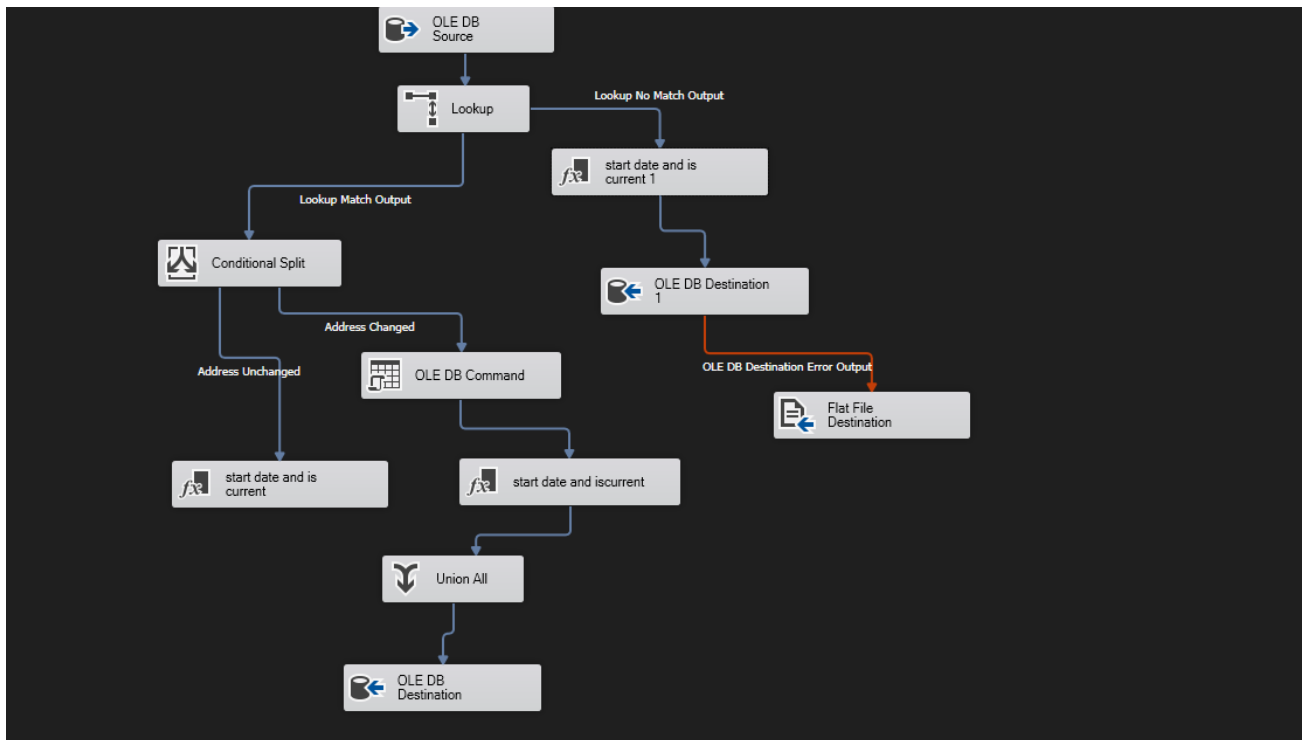
3.10 Load_DW Package — Data Flow: Dim_Category_Translation



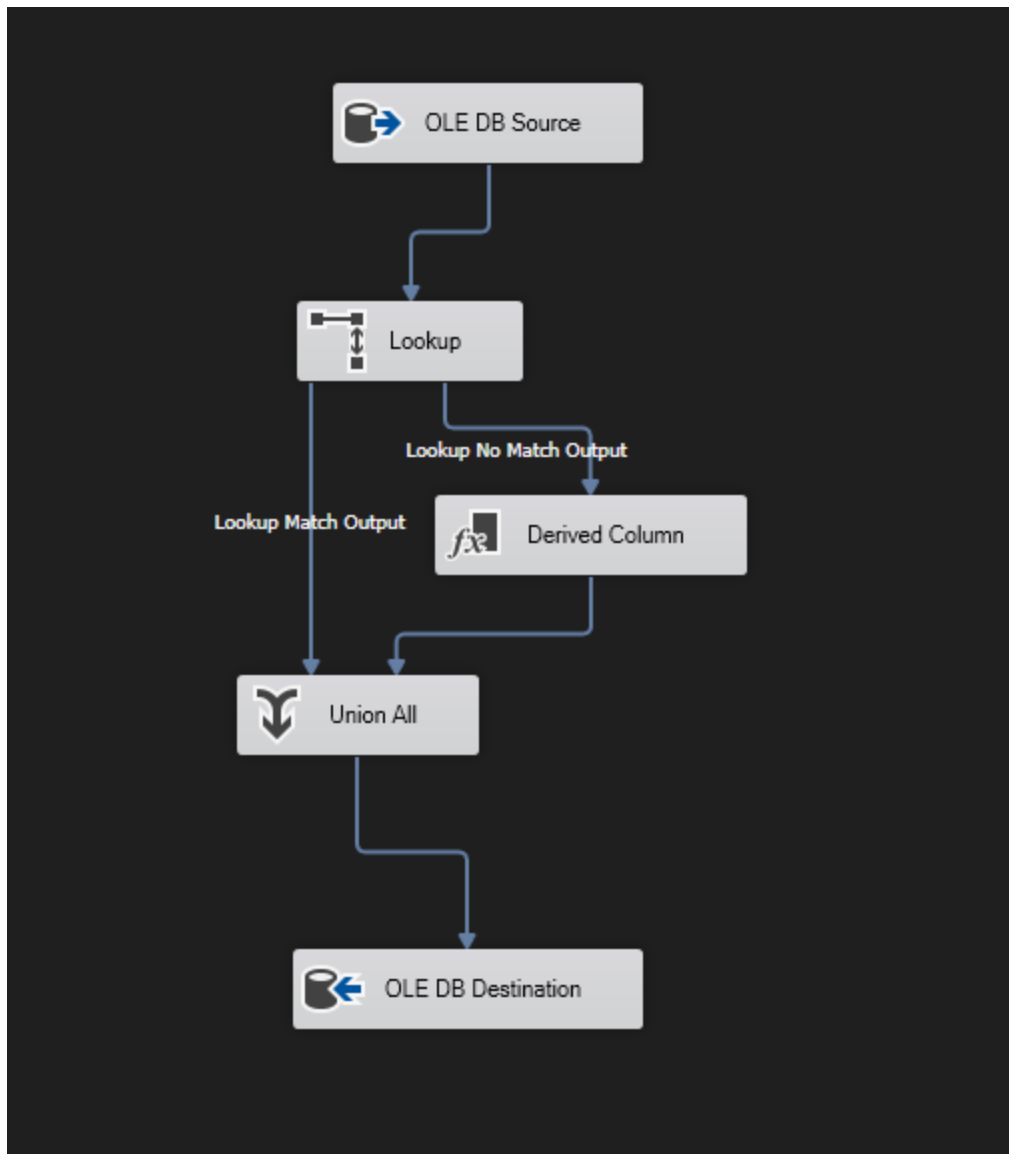
3.11 Load_DW Package — Data Flow: Dim_Geolocation



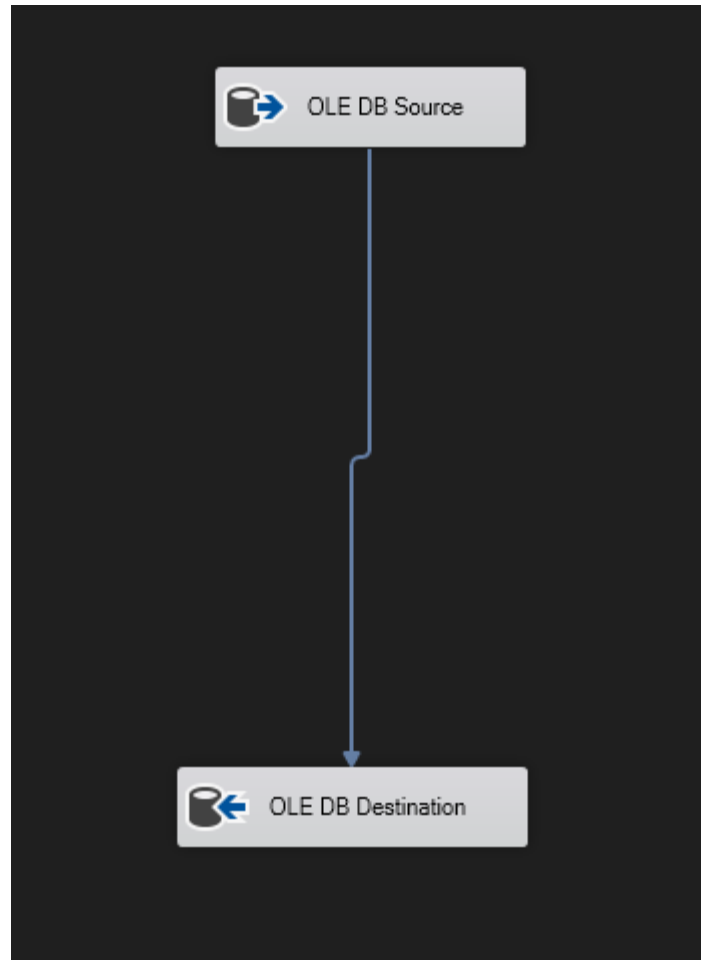
3.12 Load_DW Package — Data Flow: Dim_Customer (SCD Type 2)



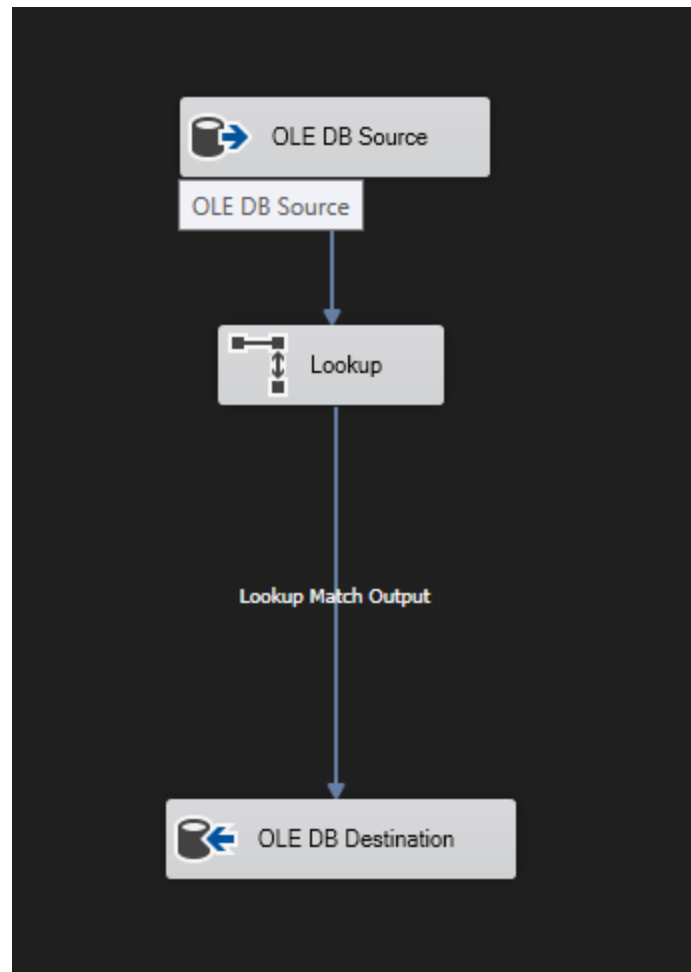
3.13 Load_DW Package — Data Flow: Dim_Product



3.14 Load_DW Package — Data Flow: Dim_Sellers



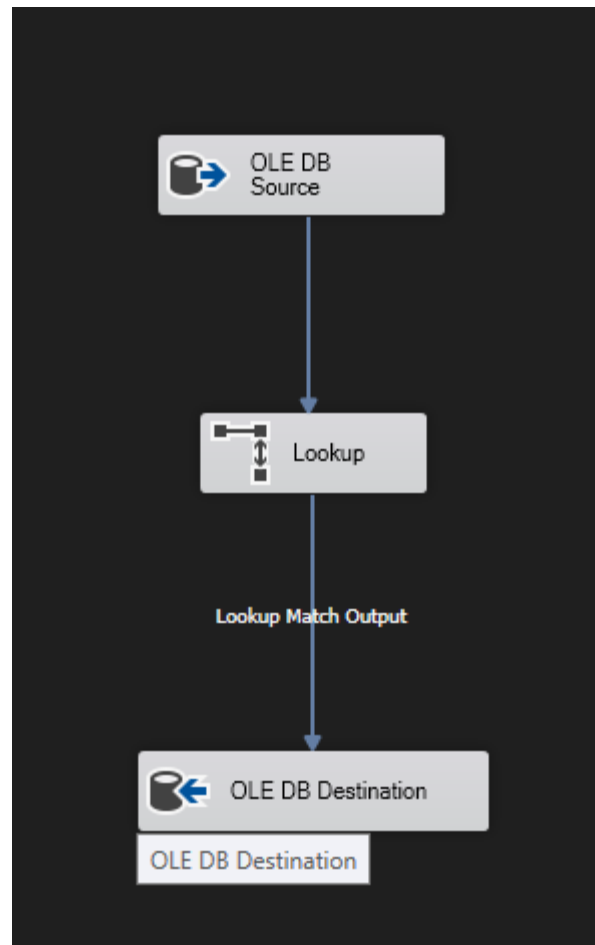
3.15 Load_DW Package — Data Flow: Dim_Orders (Incremental)



3.16 Load_DW Package — Data Flow: Fact_Order_Items



3.17 Load_DW Package — Data Flow: Fact_Payments



3.18 Load_DW Package — Data Flow: Fact_Reviews



4. Analytical Queries on Fact Tables

The following queries demonstrate the analytical capabilities of each fact table in the data warehouse.

4.1 Fact_Order_Items Queries

Query 1: Revenue by Product Category per Month

```
USE Brazilian_DW
```

```
SELECT
    YEAR(o.order_purchase_timestamp)      AS order_year,
    MONTH(o.order_purchase_timestamp)     AS order_month,
    ct.product_category_name_english      AS category,
    COUNT(DISTINCT fi.order_id)           AS total_orders,
    SUM(fi.price)                         AS total_revenue,
    SUM(fi.freight_value)                 AS total_freight,
    SUM(fi.price + fi.freight_value)      AS total_with_freight
FROM Fact_Order_Items fi
JOIN Dim_Orders o          ON fi.order_id = o.order_id
JOIN Dim_Product p        ON fi.product_id = p.product_id
JOIN Dim_Category_Translation ct ON p.product_category_name = ct.product_category_name
GROUP BY
    YEAR(o.order_purchase_timestamp),
    MONTH(o.order_purchase_timestamp),
    ct.product_category_name_english
ORDER BY order_year, order_month, total_revenue DESC;
```

Insight: This query joins the order items with the orders and product category tables. It groups the data by year, month, and category to show how much revenue each category made per month, how many orders were placed, and the total freight cost.

Results		Messages					
	order_year	order_month	category	total_orders	total_revenue	total_freight	total_with_freight
1	2016	9	health_beauty	1	134.97	8.49	143.46
2	2016	9	furniture_decor	1	72.89	63.34	136.23
3	2016	9	telephony	1	59.50	15.56	75.06
4	2016	10	furniture_decor	51	5807.89	1244.39	7052.28
5	2016	10	perfumery	30	5688.70	535.14	6223.84
6	2016	10	health_beauty	44	4552.51	940.87	5493.38
7	2016	10	toys	25	4465.09	520.99	4986.08
8	2016	10	consoles_games	10	3882.26	207.16	4089.42
9	2016	10	watches_gifts	5	3360.24	107.80	3468.04
10	2016	10	sports_leisure	19	3333.64	593.47	3927.11
11	2016	10	auto	11	1833.25	424.31	2257.56
12	2016	10	air_conditioning	5	1707.09	197.82	1904.91
13	2016	10	baby	11	1630.16	188.92	1819.08
14	2016	10	computers_accessories	18	1399.32	340.77	1740.09
15	2016	10	garden_tools	5	1359.88	105.31	1465.19
16	2016	10	electronics	2	1306.99	40.58	1347.57
17	2016	10	market_place	7	1306.19	272.99	1579.18
18	2016	10	office_furniture	6	1300.88	286.67	1587.55
19	2016	10	housewares	9	1287.07	255.47	1542.54
20	2016	10	cool_stuff	8	1111.00	216.11	1327.11
21	2016	10	fixed_telephony	5	704.88	69.57	774.45
22	2016	10	pet_shop	4	689.68	86.18	775.86
23	2016	10	fashion_bags_access...	8	508.30	148.19	656.49

4.2 Fact_Payments Queries

Fact_Payments captures how customers pay for orders. It enables analysis of preferred payment methods, installment behavior, and revenue collection timing.

Query 2: Payment Method Analysis

```
USE Brazilian_DW
```

```
SELECT
    payment_type,
    COUNT(*)                                AS total_transactions,
    COUNT(DISTINCT order_id)                AS total_orders,
    SUM(payment_value)                      AS total_revenue,
    AVG(payment_value)                      AS avg_payment_value,
    AVG(CAST(payment_installments AS FLOAT)) AS avg_installments,
    MAX(payment_installments)               AS max_installments
FROM Fact_Payments
GROUP BY payment_type
ORDER BY total_revenue DESC;
```

Insight: This query groups all payments by payment type to show how many transactions each method had, the total and average payment value, and the average number of installments used per payment method.

	payment_type	total_transactions	total_orders	total_revenue	avg_payment_value	avg_installments	max_installments
1	credit_card	76795	76505	12542084.19	163.319020	3.50715541376392	24
2	boleto	19784	19784	2869361.27	145.034435	1	1
3	voucher	5775	3866	379436.87	65.703354	1	1
4	debit_card	1529	1528	217989.79	142.570170	1	1
5	not_defined	3	3	0.00	0.000000	1	1

4.3 Fact_Reviews Queries

Fact_Reviews captures customer satisfaction after order delivery. It enables analysis of service quality over time, by seller, and by product category.

Query 3: Review Score by Seller

USE Brazilian_DW

```

SELECT
    s.seller_id,
    s.seller_city,
    COUNT(r.fact_review_sk) AS total_reviews,
    AVG(CAST(r.review_score AS FLOAT)) AS avg_review_score,
    SUM(CASE WHEN r.review_score = 5 THEN 1 ELSE 0 END) AS five_star_reviews,
    SUM(CASE WHEN r.review_score <= 2 THEN 1 ELSE 0 END) AS bad_reviews
FROM Fact_Reviews r
JOIN Dim_Orders o ON r.order_id = o.order_id
JOIN Fact_Order_Items fi ON o.order_id = fi.order_id
JOIN Dim_Sellers s ON fi.seller_id = s.seller_id
GROUP BY s.seller_id, s.seller_city
HAVING COUNT(r.fact_review_sk) >= 10
ORDER BY avg_review_score DESC;

```

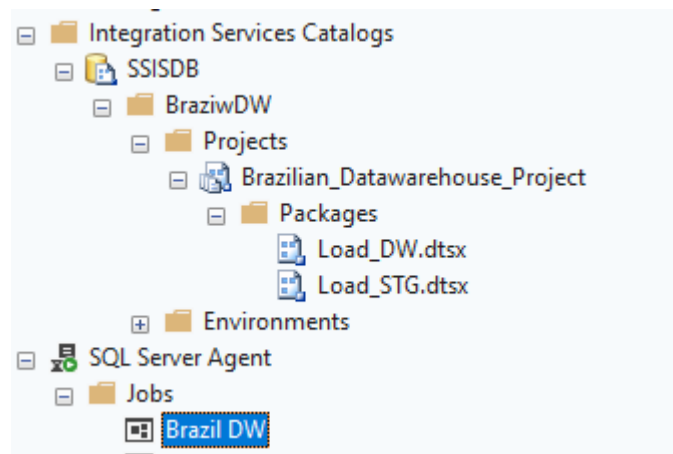
Insight: This query links reviews to orders, then to order items and sellers. It groups by seller to show how many reviews each seller received, their average score, how many 5 star reviews they got, and how many bad reviews (score 2 or below).

	seller_id	seller_city	total_reviews	avg_review_score	five_star_reviews	bad_reviews
1	0570350b23eda5444f8d1d9544932058	sao paulo	11	5	11	0
2	a08692680c77d30a0b4280da5df01c5a	sao paulo	17	5	17	0
3	2addf05f476d0637864454e93ba673d5	brasilia	12	5	12	0
4	c8c1bea22194a4eefa2dc9a9fa89f536	porto belo	13	5	13	0
5	0b36063d5818f81ccb94b54adfaebbf5	jaragua do sul	15	5	15	0
6	53b0300ca793f9834cd69c0678d35ee8	indaiatuba	10	5	10	0
7	404e1ba01358af4cd63f679b2c4d1fa1	juiz de fora	12	5	12	0
8	b2eef5ea250510da76590ca79d60e5d	sao paulo	11	5	11	0
9	48efc9d94a9834137efd9ea76b065a38	curitiba	34	5	34	0
10	ab27bbbad5239bc31a34709275a70db4	tatui	11	5	11	0
11	334cab711dee080b079fa5779b584783	santo andre	26	4.96153846153846	25	0
12	41c2bad7229b0c25e6becf179ebf63ff	natal	23	4.95652173913043	22	0
13	b1fecf4da1fa2689bccffa0121953643	campo grande	20	4.95	19	0
14	9d681c7e12db302cb261e721040dde65	curitiba	14	4.92857142857143	13	0
15	2d518637f53161b973e01f56ea4bb88e	sao paulo	13	4.92307692307692	12	0
16	d4910f1cdcfdbd48d6b316e395d4a23	santo andre	12	4.91666666666667	11	0
17	2f4b0d3b9634b647d4edee577d7ebb7e	sao paulo	11	4.90909090909091	10	0
18	788e857f317e53de488d393e65a80f45	sao paulo	11	4.90909090909091	10	0
19	5c030029b5916fed0986310385ec9009	sao jose	10	4.9	9	0
20	f44b6f775a2b390380e0e30b5c9d3eee	sao paulo	10	4.9	9	0
21	5b2e94f2c22b39ccc24d34ad0f669672	sao paulo	10	4.9	9	0
22	2d2322d842118867781fc737e96d59a1	sao paulo	10	4.9	9	0

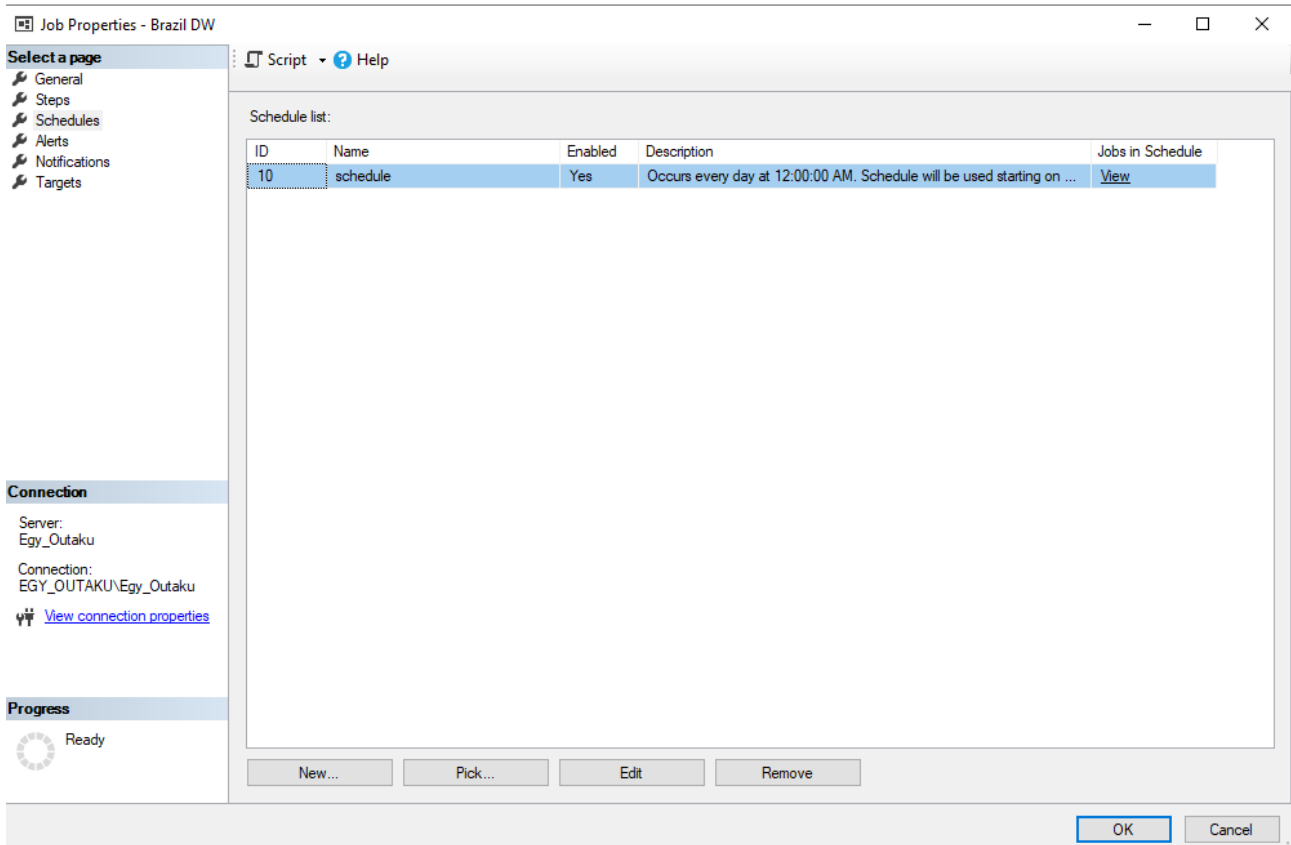
5. Deployed SSIS Packages and Schedule

Both SSIS packages were deployed to the SSIS Catalog (SSISDB) on SQL Server and scheduled using SQL Server Agent.

5.1 Deployed Packages in SSIS Catalog



5.2 SQL Server Agent Job — Schedule

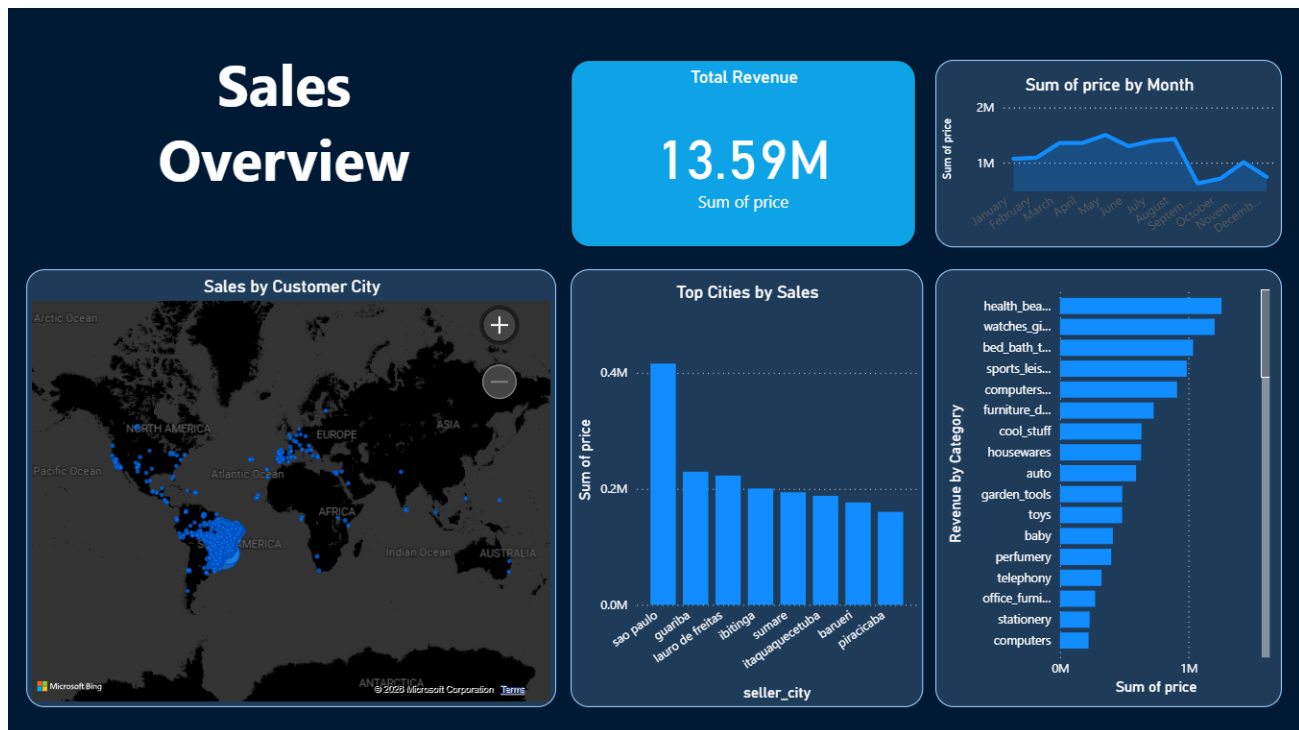


This packages is scheduled to run daily at 12:00 am to execute.

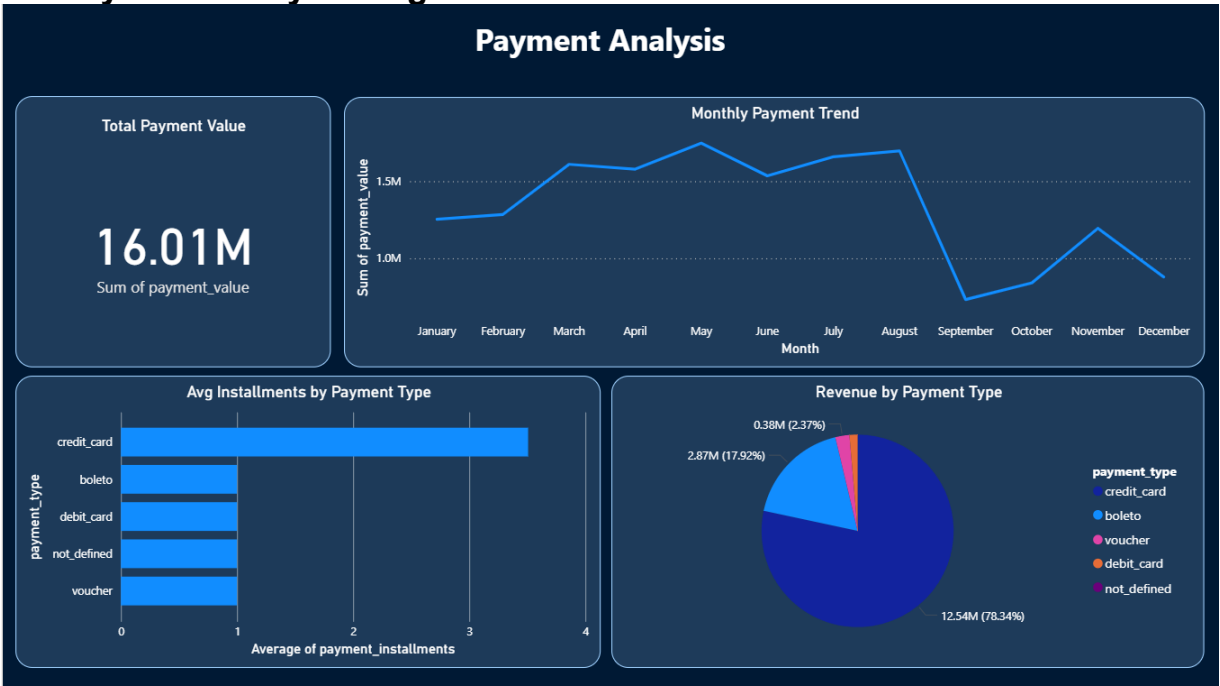
6. [Bonus] Interactive Power BI Dashboard

An interactive dashboard was built in Microsoft Power BI Desktop, connected directly to the Brazilian_DW SQL Server database. The dashboard provides visual analytics across the three fact tables and their dimensions.

6.1 Sales Overview Page



6.2 Payment Analysis Page



6.3 Customer Satisfaction (Reviews) Page



The dashboards include the following key visuals:

- KPI cards: Total Payment Value, Average Review Score

- Monthly payment trend (line chart)
- Revenue by payment type (pie chart)
- Average installments by payment type (horizontal bar chart)
- Average review score over time (line chart)
- Review score distribution (horizontal bar chart)
- Geographic map of average review scores by customer state
- Dark-themed analytical dashboard layout with interactive visual segmentation and category-based insights

All visuals support cross-filtering — clicking any element filters the entire report page, enabling drill-down analysis.